

Chapter 14 Homework

Chapter 14 Homework – Matrix Decompositions

Part A – Python Skills Reinforcement

Problem 1 – LU Decomposition

Given:

```
A =  
[  
  [2,1,1],  
  [4,-6,0],  
  [-2,7,2]  
]
```

Tasks:

1. Compute P, L, and U using `scipy.linalg.lu()`.
2. Print all matrices.
3. Verify that $P @ L @ U$ reconstructs A.

Problem 2 – QR Decomposition

Given:

```
A =  
[  
  [1,2],  
  [3,4],  
  [5,6]  
]
```

Tasks:

1. Compute Q and R.
2. Print Q.
3. Print R.
4. Reconstruct A using $Q @ R$.
5. Verify reconstruction error is near zero.

Problem 3 – Cholesky Decomposition

Given:

```
A =  
[  
  [4,2,2],  
  [2,5,1],  
  [2,1,3]  
]
```

Tasks:

1. Compute the Cholesky decomposition.
2. Print L.

3. Reconstruct A using $L @ L.T$.
4. Explain why A must be symmetric.

Problem 4 – Compare Shapes

For LU, QR, and Cholesky:

1. Print shapes of all factors.
2. Explain why the shapes differ.
3. Explain when each decomposition is applicable.

Part B – Practical Problem: Sensor Correlation Compression

Assume:

```
A =  
[  
  [12,15,18],  
  [14,17,22],  
  [16,20,25]  
]
```

Columns represent:
Radar, EO/IR, Acoustic sensors.

Step 1

Compute:

$\text{rank}(A)$

$\text{det}(A)$

Interpret the results.

Step 2

Perform QR decomposition.

Print:

Q

R

Interpret:

- What information is contained in Q?
- What information is contained in R?

Step 3

Reconstruct the matrix.

Compute:

$A_{\text{reconstructed}} = Q @ R$

Verify the reconstruction error is approximately zero.

Step 4

Suppose the matrix grows to:

1000 sensors x 5000 observations

Explain why factorization methods are preferable to directly computing inverses.

Step 5 – Research Question

Write a short paragraph explaining:

'Why matrix decomposition is considered one of the most important tools in numerical linear algebra.'

Discuss:

- Numerical stability
- Computational efficiency
- Solving least-squares problems
- Why decompositions appear throughout machine learning